

CHAPTER III

Preferences and Investor Choice

Utility functions, iso-utility curves, the safety-first criterion, the Sharpe Ratio, and Value at Risk.

Introduction

The previous chapter presented Markowitz's portfolio selection model while omitting a crucial element: how individual investors actually *choose* among portfolios. Although the efficient frontier represents the superior combination of assets, it contains infinitely many possibilities. The central challenge becomes selecting one optimal portfolio from this set, requiring investors to express their preferences within risk-return space.

Investors make decisions based on numerous factors that cannot be reduced to two dimensions. Furthermore, real-world investing allows strategy adjustments as circumstances change — a flexibility the single-period Markowitz model doesn't accommodate. This chapter explores methods for approximating investor preferences in mean-variance space, acknowledging their limitations when applied to multi-period decision-making.

I. Choosing a Single Portfolio

Iso-Utility Curves and Investor Preferences

One approach involves mathematical modeling of investor preferences through **iso-utility curves**. These curves function like topographic map contours, representing consistent utility levels. They demonstrate how investors trade off risk against return in two-dimensional space. Like elevation lines on maps, iso-utility curves are nested, with higher curves representing greater utility levels.

The mathematical foundation typically uses convenient functions — logarithmic or quadratic — to capture preference structures. The essential characteristic requires increasing return demands as risk exposure increases. A common form:

UTILITY FUNCTION

← Investment Theory

$$U = \bar{R}_p - \frac{1}{2} \lambda \sigma_p^2$$

where λ is the investor's risk-aversion coefficient. Higher λ means more risk aversion.

Characterizing Risk Aversion

Iso-utility curve curvature reveals investor risk preferences. Four investor types exhibit distinct curve patterns:

1. **More risk-averse investors:** Demand sharply higher expected returns when volatility increases (steep curves)
2. **Moderately risk-averse investors:** Show balanced return-risk requirements
3. **Less risk-averse investors:** Display flatter curves with moderate return increases for additional risk
4. **Risk-loving investors:** Paradoxically demand *lower* expected returns while assuming greater risk to maintain utility levels

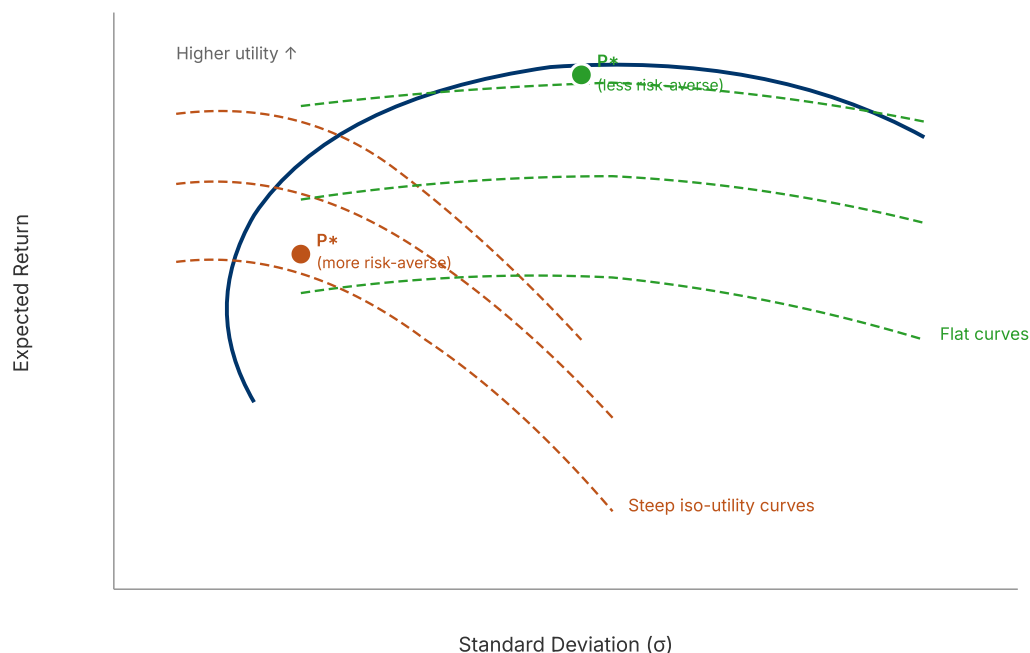


Figure 3.1. Iso-utility curves and optimal portfolio selection. More risk-averse investors (steep curves, red) select portfolios nearer the minimum-variance end. Less risk-averse investors (flatter curves, green) select portfolios further along the frontier, accepting more risk for higher expected return.

Finding Optimal Portfolios

The optimization procedure identifies where the efficient frontier is **tangent** to the highest achievable iso-utility curve for that investor. This tangency point represents the unique

portfolio providing maximum utility for risk-averse individuals.

← Investment Theory

Practical Limitations

The methodology's significant drawback involves determining individual or institutional risk aversion. Asset allocators must map client preferences — analogous to mapping unknown terrain — without certainty regarding consistency over time. Preferences may shift unpredictably between periods.

II. Another Approach: Preferences about Distributions

The Shortfall Criterion

An alternative strategy focuses on return distribution characteristics, specifically probability mass in lower-tail regions. This **"shortfall" criterion** offers simplicity by avoiding complete preference mapping. Instead, investors specify a *floor return* — a minimum threshold they want to avoid breaching. The criterion then selects the efficient frontier portfolio minimizing the probability of returns dropping below that floor.

Mathematical Expression

For a specified floor return equal to the riskless rate R_f , calculate this ratio for each frontier portfolio:

SHORTFALL RATIO

$$S = \frac{\bar{R}_p - R_f}{\sigma_p}$$

The shortfall ratio resembles a t -statistic; higher values indicate greater probability of exceeding the floor. The portfolio maximizing this value has the highest probability of surpassing R_f .

Graphical Solution

The process can be solved graphically without complex calculations:

1. Identify the floor return level on the vertical axis
2. Draw a line from this point tangent to the efficient frontier
3. This tangency point minimizes the probability of returns falling below the specified floor

The Sharpe Ratio

← Investment Theory

When R_f (the riskless rate) serves as the floor, the tangency line's slope equals the Sharpe Ratio:

SHARPE RATIO

$$\text{Sharpe Ratio} = \frac{\bar{R}_p - R_f}{\sigma_p}$$

The portfolio maximizing this ratio possesses special significance: it minimizes the probability of underperforming Treasury bills economy-wide. Equivalently, it maximizes the probability of delivering an equity premium — the single best portfolio for beating T-bills if forced to choose one.

Key Concept: The Sharpe Ratio

The Sharpe Ratio measures return per unit of risk relative to the risk-free rate. The portfolio maximizing it is the **tangency portfolio** — the same portfolio M that defines the Capital Market Line. It is the single best risky portfolio for any investor who can borrow or lend at the risk-free rate.

Flexibility of Safety-First

The "safety-first" framework accommodates multiple approaches:

1. **Maximizing probability of exceeding a floor:** Find the tangency point as described above
2. **Testing feasibility:** Specify both a desired floor and probability; determine whether portfolios meeting these criteria exist
3. **Finding sustainable floors:** Pose the inverse question — "Given my acceptable probability, what floor can I reliably achieve?"

For the third approach, set the slope equal to appropriate t -statistic values:

CONFIDENCE LEVEL	REQUIRED SLOPE (T -VALUE)
90%	1.28
95%	1.644
99%	2.32

Once the slope is chosen, move the line vertically until tangent to the efficient frontier. This yields both a portfolio choice and its corresponding floor return.

III. A Note on Value at Risk

← Investment Theory

Definition and Purpose

Value-at-Risk (VaR) measures potential portfolio losses over specified time horizons. It has become increasingly prevalent in banking (for capital requirements calculation) and investment management (for risk control). VaR provides a probability-based loss estimate given a particular confidence interval and time period.

Calculation Method

Assuming normally distributed portfolio returns, VaR calculation requires the portfolio expected return $\bar{R}_p$, portfolio standard deviation σ_p , and the t -statistic T for the chosen confidence interval:

VAR RETURN THRESHOLD

$$R_{\text{VaR}} = \bar{R}_p - T \cdot \sigma_p$$

VALUE AT RISK (DOLLARS)

$$\text{VaR} = |R_{\text{VaR}}| \times \text{Total Portfolio Value}$$

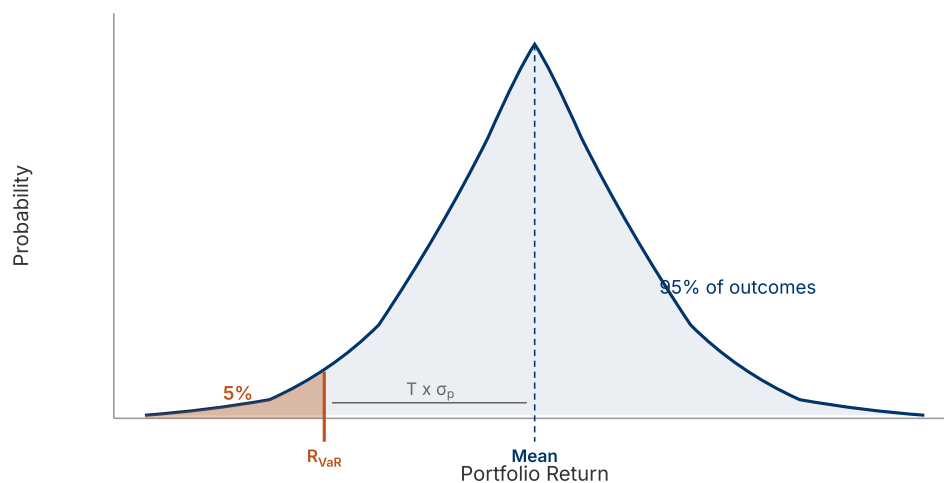


Figure 3.2. Value at Risk (VaR) at the 95% confidence level. The shaded left tail represents the 5% probability of losses exceeding the VaR threshold. $R_{\text{VaR}} = \bar{R}_p - T \cdot \sigma_p$.

Practical Example

← Investment Theory

Consider a \$100 million pension portfolio with 60% stocks and 40% bonds, analyzed at the 95% confidence interval over one month.

Assumptions:

- Monthly expected stock return: 1%
- Monthly expected bond return: 0.7%
- Stock standard deviation: 5%
- Bond standard deviation: 3%
- Stock-bond correlation: 0.5
- 95% confidence t -statistic: 1.64

Portfolio Expected Return:

$$\bar{R}_p = (0.6)(0.01) + (0.4)(0.007) = 0.0088 \text{ (0.88\%)}$$

Portfolio Standard Deviation:

$$\sigma_p = \sqrt{(0.6)^2(0.05)^2 + (0.4)^2(0.03)^2 + 2(0.5)(0.6)(0.4)(0.05)(0.03)} = 0.038$$

Value-at-Risk:

$$R_{\text{VaR}} = 0.0088 - 1.64 \times 0.038 = -0.054 \text{ (-5.4\%)}$$

$$\text{Monthly VaR} = \$100,000,000 \times 0.054 = \$5,400,000$$

The portfolio is expected to experience losses of at least

5.4 million approximately one month out of every twenty, though this calculation does not suggest that 94.6 million escapes risk entirely.

Key Concept: Value at Risk

VaR expresses portfolio risk as a dollar amount: "With X% confidence, we will not lose more than \$Y over the next period." It is a direct extension of the safety-first framework and requires the same normality assumptions. VaR is widely used in banking regulation and institutional risk management.

Key Assumptions and Limitations

← Investment Theory

1. **Normal distribution of returns:** This assumption frequently breaks down when portfolios contain derivatives, producing fat-tailed distributions

2. **Historical representativeness:** Estimated distributions and correlations must accurately reflect future conditions; estimation error increases significantly with numerous asset classes
3. **No return autocorrelation:** The model assumes independence across periods; positive trends cause losses to compound across multiple periods

IV. Conclusion

Creating efficient frontiers from historical or projected statistics involves inherent uncertainty due to statistical estimation errors. However, this uncertainty is minor compared to the difficulty of translating investor preferences into mean-standard deviation space. Economic knowledge about preferences remains limited, particularly within single-period frameworks. While economists understand that people prefer more to less and typically avoid uncompensated risk, preferences beyond these basics involve considerable speculation. Consistency across time periods remains questionable.

The theoretical solution requires specifying an investor utility function, deriving indifference curves, and locating the highest attainable utility level within the feasible set — typically a tangency point. Practical implementation proves difficult: estimating utility functions and subsequently explaining them to investors presents major obstacles.

The "**safety-first**" approach offers a practical alternative beginning with a straightforward preference question: "What is your floor return?" Once specified, portfolio selection follows logically. This methodology allows identifying the probability of exceeding the selected floor by observing the tangency line's slope. The framework further permits portfolio identification by simultaneously specifying both a floor and probability.

Value-at-Risk represents an increasingly popular risk measurement and control method, functioning as a straightforward extension of safety-first principles when portfolio assets display normally distributed returns.

Epilogue: Simplification Through Risk-Free Securities

Introducing a genuine risk-free security dramatically simplifies portfolio decisions for all investors. The optimal choice reduces to selecting proportions of the riskless asset and a single risky portfolio (T) — the tangency portfolio:

- **More risk-averse (MRA) investors:** Hold combinations of the tangency portfolio and Treasury bills

← Investment Theory

- Less risk-averse (LRA) investors: Borrow at the riskless rate and invest proceeds entirely in the tangency portfolio

If practitioners could definitively identify the tangency portfolio's composition, a single product could serve every investor's needs by simply varying proportions of the riskless asset and this portfolio.

What constitutes portfolio T? The answer is provided in the subsequent chapter.

Further reading: For utility function approaches to risk, see Campbell Harvey's resource on [Optimal Portfolios](#). For a comprehensive hypertext investment decision-making resource, see William Sharpe's *Macro-Investment Analysis*.

Interactive Tool: Explore these concepts interactively using the [Rice Business Investments Compendium](#), created by Kerry Back and Kevin Crotty at the Jones Graduate School of Business, Rice University.

Acknowledgement: The interactive tools referenced in this textbook are from the [Rice Business Investments Compendium](#), created and maintained by Kerry Back and Kevin Crotty at the Jones Graduate School of Business, Rice University. We gratefully acknowledge their contribution to investment education.

← Chapter II: The Efficient Frontier

Chapter IV: Portfolio Approach to Risk →